

- 4 (a) Cells of the immune system have cell surface receptors that detect molecules made by pathogens. One of these cell surface receptors is known as TLR8.

The gene *TLR8* is found on the X chromosome in humans.

Fig. 4.1 shows the production of messenger RNA (mRNA) formed from the gene *TLR8* in the nucleus of a macrophage.

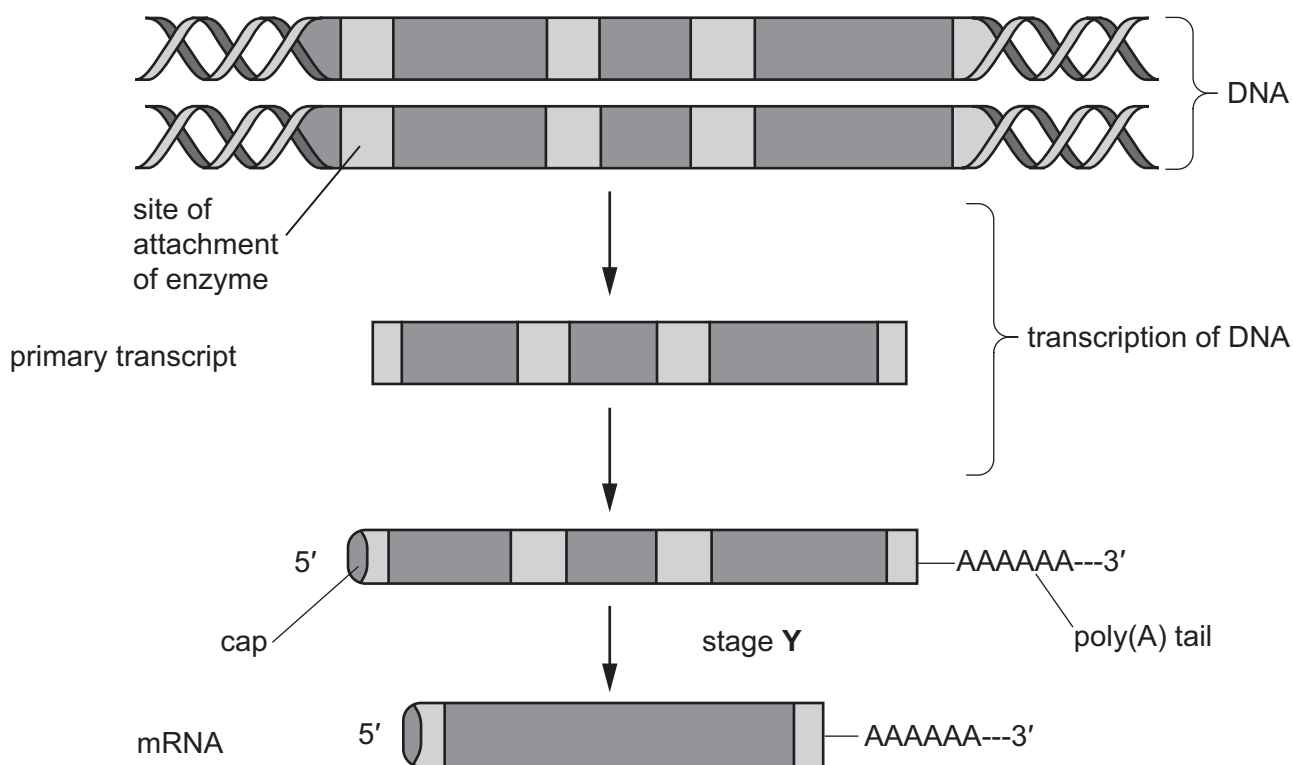


Fig. 4.1

- (i) Name the enzyme that catalyses the transcription of DNA.

..... [1]

- (ii) Fig. 4.1 shows that the primary transcript is modified by the addition of nucleotides to both ends of the molecule.

The cap shown in Fig. 4.1 is a guanine nucleotide that is added to the 5' end of RNA. The poly(A) tail added to the 3' end consists of many adenine nucleotides. The cap and the tail have a function in stage Y and are also important for the stability and role of mRNA.

Suggest the functions of the cap and the poly(A) tail in the stability and role of mRNA.

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(iii) Describe the process that occurs at stage Y in Fig. 4.1.

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..... [3]

(b) The nucleotide sequence TTAGGG is repeated in the direction 5' to 3' in the telomeres of human chromosomes.

(i) State where in a chromosome the telomeres are found.

..... [1]

(ii) Outline the role of telomeres.

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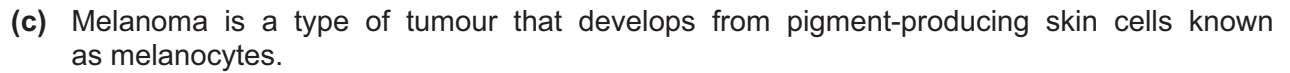
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..... [3]



[4]

- T-vec is a new drug that has been developed to treat melanoma that has spread to other parts of the body, including lymph nodes.

Explain the effects of the cytokines and antigens released from melanoma cells in stimulating the immune system to destroy the cancerous cells.

..... [5]

[Total: 19]





Question 5 starts on page 16.